

Frontend theory assignment 2

1. What is CSS Box Model?

CSS Box Model is used to describe how every HTML element is treated as a box in CSS.

It consists of four parts:

1. **Content** – The actual text or image inside the element.
2. **Padding** – Space between the content and border.
3. **Border** – A line around the padding and content.
4. **Margin** – Space outside the border.

Example:

```
div {  
  width: 200px;  
  padding: 20px;  
  border: 2px solid black;  
  margin: 10px;  
}
```

So, the order is:

Content → Padding → Border → Margin

2. What is Media Query? Explain with a sample code.

A media query is used in CSS to make a webpage responsive according to the screen size or device.

For example, we can change the layout when the screen width becomes 600px or less.

```
.container {  
  
  display: flex;  
  
}
```

```
@media (max-width: 600px) {  
  
  .container {  
  
    flex-direction: column;  
  
  }  
}
```

```
}  
  
}
```

Here, when the screen width is 600px or less, the items are displayed vertically instead of horizontally.

3. What are the features of Bootstrap?

Bootstrap is a popular CSS framework used to create responsive and attractive websites easily.

Main features of Bootstrap are:

1. **Responsive design** – Websites work on different screen sizes.
2. **Grid system** – Helps to create page layouts using rows and columns.
3. **Predefined CSS classes** – Provides ready-made classes for styling.
4. **Components** – Provides buttons, cards, navbar, forms, alerts, etc.
5. **Easy to use** – Less CSS code is required.
6. **Mobile-first** – Websites are designed to work properly on mobile devices.

4. Write an example of basic grid structure in Bootstrap.

Bootstrap grid system divides a webpage into **12 columns**. We can use rows and columns to create the layout.

Example:

```
<div class="container">  
  
  <div class="row">  
  
    <div class="col-md-4">Column 1</div>  
  
    <div class="col-md-4">Column 2</div>  
  
    <div class="col-md-4">Column 3</div>  
  
  </div>  
  
</div>
```

Here, the three columns take $4 + 4 + 4 = 12$ columns.

So, the complete row is divided into three equal parts.

5. What are CSS Selectors?

CSS selectors are used to select HTML elements on which we want to apply CSS styles.

Some common CSS selectors are:

1. Element selector – Selects an HTML element.

```
p {  
  color: blue;  
}
```

2. Class selector – Selects elements using a class.

```
.box {  
  color: red;  
}
```

3. ID selector – Selects an element using its ID.

```
#heading {  
  color: green;  
}
```

4. Universal selector – Selects all elements.

```
* {  
  margin: 0;  
}
```

6. What are CSS backgrounds? List the properties.

CSS background properties are used to set the background of an HTML element.

Main background properties are:

1. background-color – Sets the background color.
2. background-image – Sets an image as the background.

3. background-repeat – Controls whether the background image repeats.
4. background-position – Sets the position of the background image.
5. background-size – Sets the size of the background image.
6. background-attachment – Controls whether the background scrolls or remains fixed.

Example:

```
body {  
  
    background-color: lightblue;  
  
    background-image: url("image.jpg");  
  
    background-size: cover;  
  
    background-position: center;  
  
}
```

7. How can we overlap elements in CSS?

Elements can be overlapped in CSS by using positioning and the z-index property.

position: absolute can be used to place one element over another element.

Example:

```
.box1 {  
  
    position: absolute;  
  
    top: 50px;  
  
    left: 50px;  
  
}
```

```
.box2 {  
  
    position: absolute;  
  
    top: 80px;  
  
    left: 80px;  
  
    z-index: 2;  
  
}
```

Here, both elements can overlap and the element with a higher z-index appears on top.

8. What are the different positioning properties in CSS?

CSS provides different position properties to control the position of elements.

1. **static** – Default position of an element.
2. **relative** – Element is positioned relative to its normal position.
3. **absolute** – Element is positioned relative to its nearest positioned ancestor.
4. **fixed** – Element stays fixed at the same position even when the page is scrolled.
5. **sticky** – Element behaves like relative until a particular scroll position and then behaves like fixed.

Example:

```
.box {  
  
    position: relative;  
  
    top: 20px;  
  
    left: 10px;  
  
}
```

9. What is ECMAScript?

ECMAScript is a standard or specification that defines how scripting languages like JavaScript should work.

JavaScript follows the ECMAScript standard.

ECMAScript defines things such as:

- Variables
- Functions
- Data types
- Operators
- Loops
- Classes
- Objects

Different versions of ECMAScript introduce new features to JavaScript.

10. What is the difference between var, let and const in JavaScript? Explain Temporal Dead Zone.

The main differences between `var`, `let` and `const` are:

Feature	var	let	const
Scope	Function scope	Block scope	Block scope
Redeclaration	Allowed	Not allowed	Not allowed
Reassignment	Allowed	Allowed	Not allowed
Hoisting	Yes	Yes, but TDZ	Yes, but TDZ

Example:

```
var a = 10;
```

```
let b = 20;
```

```
const c = 30;
```

```
a = 15; // allowed
```

```
b = 25; // allowed
```

```
// c = 35; // not allowed
```

Temporal Dead Zone (TDZ):

Temporal Dead Zone is the time between entering a block and the declaration of a `let` or `const` variable. During this time, the variable cannot be accessed.

Example:

```
console.log(x); // Error
```

```
let x = 10;
```

Here, x is in the Temporal Dead Zone until the `let x` declaration is reached.